LEARNING WHAT TO CHANGE

From Visual Reasoning to Scientific Hypothesis Revision

Chimdi Walter Ndubuisi · University of Missouri REVIEW DRAFT 19 October: for collaborator review, not for print

Can a reasoner recognize when its explanation is inadequate, and determine what needs to change?

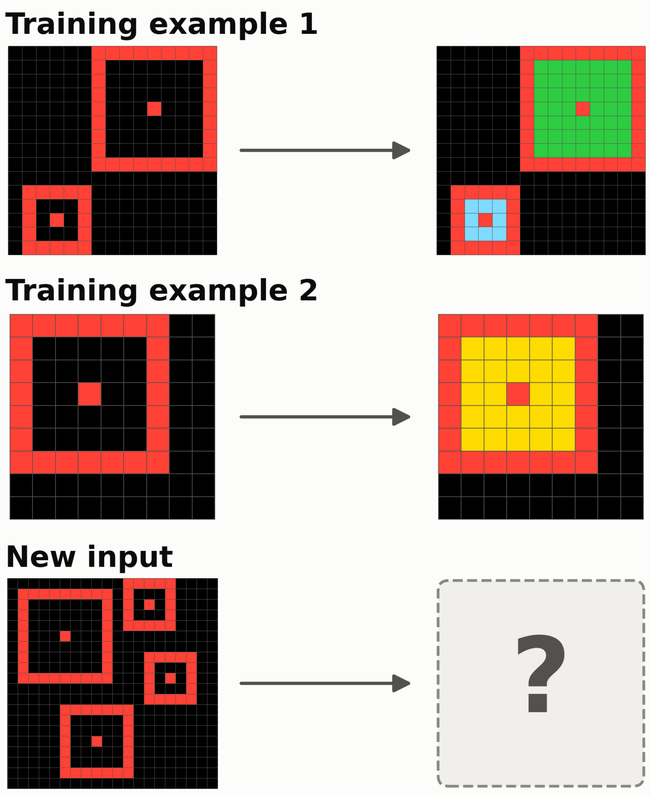
When an explanation is inadequate, how can a reasoner discover what is missing, choose what to revise, and test whether the revision helps? ARC-AGI-2 is the controlled testbed. A question shaped in discussions with Toni Kazic.

Aims, not yet all achieved: find objects and relations → write executable hypotheses → notice inadequacy → choose a justified revision → test it independently → transfer the procedure.

1 · Why reasoning fails

Find one rule that fits both examples, then apply it to the new input.

REAL ARC-AGI-2 TRAINING TASK 00dbd492 (public)



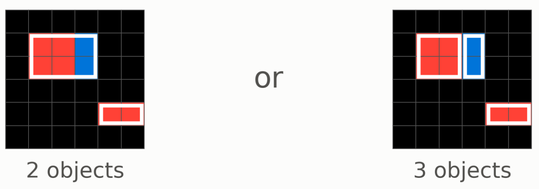
The rule here: each frame's fill colour depends on its size.

Three ways an explanation fails

ILLUSTRATIVE — drawn to explain, not a CORA result

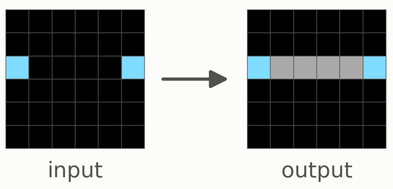
Wrong objects

The grid is split into the wrong pieces.



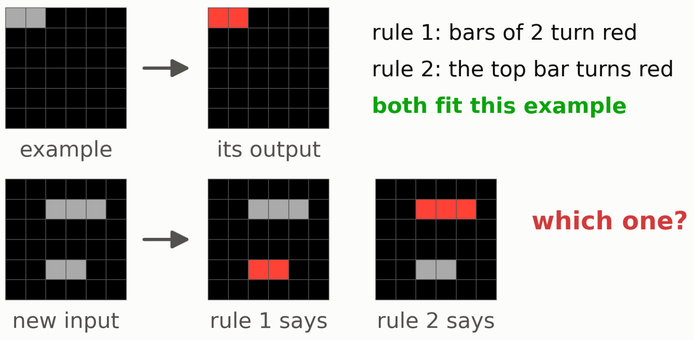
Wrong transformation

Right pieces, but no operation makes the change.



Not enough evidence

Two rules fit the example but give different answers.



Telling these apart from the examples alone is the research problem.

2 · What CORA has demonstrated

CORA writes rules as programs and checks them.

Fitting is not enough [DONE]

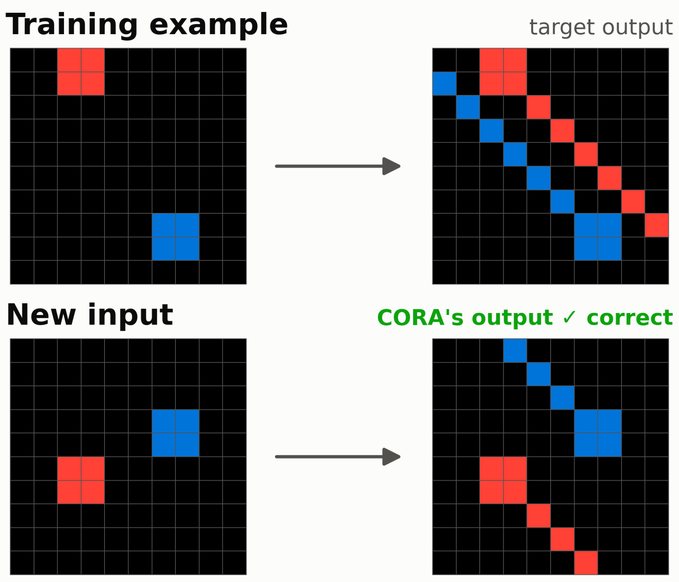
6 answers accepted, all correct

26 tasks where every rule that fit the examples failed the test

60 exposed ARC-AGI-2 training tasks; CORA's object reasoner as submitted.

A rule CORA built [DONE, NARROW]

REAL ARC-AGI-2 TRAINING TASK ka95aa62f67d9 · recorded run



The rule: each object extends a diagonal line, chosen by its colour: **blue** up-left, **red** down-right.

- ✓ CORA constructed this executable rule
- ✓ correct prediction on this development task
- ✓ rebuilt 3 of 3 times with one example left out
- = a version blind to failures found the same rule

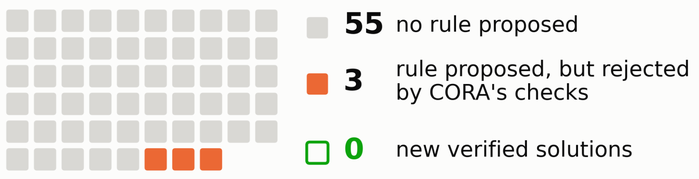
An exposed ARC training task, not an unseen Kaggle success.

Did it generalize? [NEGATIVE]

0 / 58

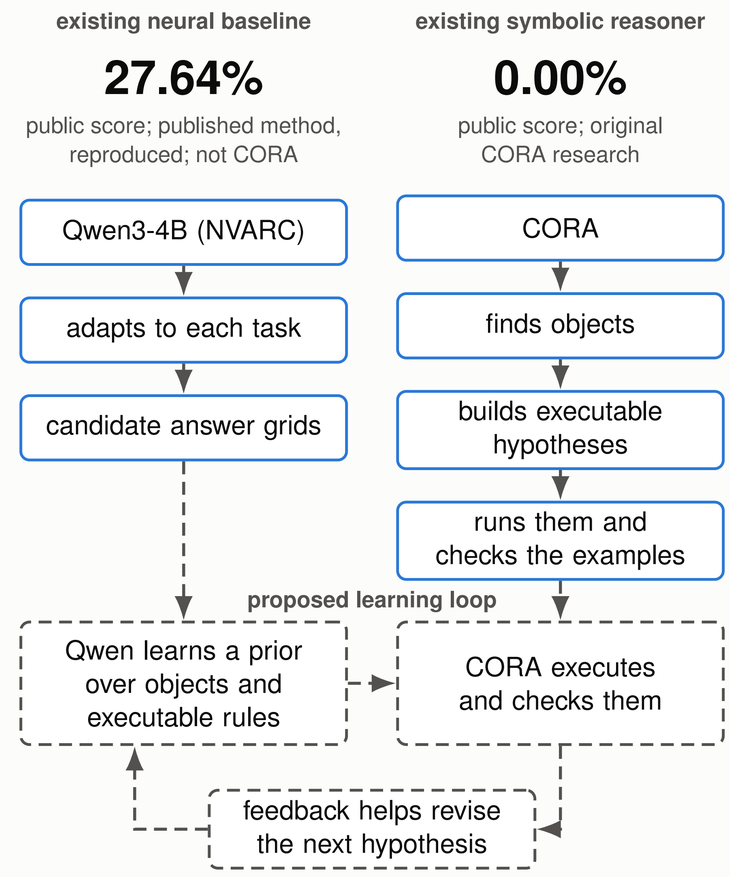
new verified solutions from rule construction

REAL · 58 separately selected, exposed training tasks



Shown: construction that used the failure record. Every variant found 0; no wrong answers.

3 · How neural learning may help



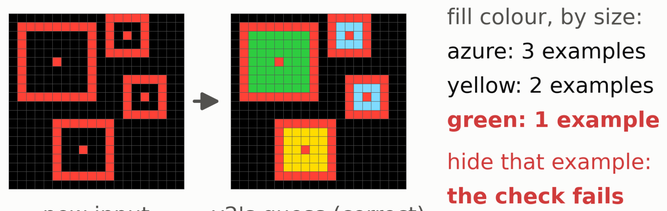
Can neural learning help a reasoner discover useful representations and rules—and can executable verification help it determine what to change when those explanations are inadequate?

Solid: existing. Dashed: proposed. The neural proposal interface has not yet been implemented or experimentally validated; the neural model outputs answer grids only.

CORA-v2 [DEVELOPMENT]

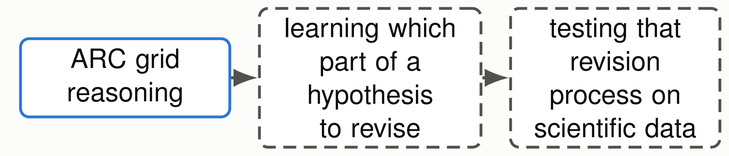
Compares rival rules; records why it is unsure.

REAL TASK 00dbd492 · CORA-v2 development run
CORRECT DEVELOPMENT PREDICTION — NOT CERTIFIED



On 118 exposed development tasks: 4 correct and 1 wrong exploratory guesses; nothing certified.

From ARC to science [FUTURE, NOT TESTED]



Possible targets: proteins, omics data, plant phenotypes, maize lesion-mimic pathways. What could transfer is the revision policy, not ARC operations.

Building a rule is not the same as learning which revision improves an unseen prediction. That is what we are trying to test.

existing, tested · proposed, not yet validated · [DONE] [NEGATIVE] [DEVELOPMENT] [FUTURE, NOT TESTED]

Related work (explainer, section 9): Popper (Cropper & Morel 2021); DreamCoder (Ellis et al. 2021); LILO (Grand et al. 2024); Model Discovery Agent (Murphy 2026). The failure-to-revision idea is not new; the question is which component to revise, against matched controls. Neural baseline: the NVARC team's published method and Qwen3-4B checkpoint, reproduced. A separate CORA experiment on inventing new operations is running (explainer, section 7). Numbers: `records/CORA_SUMMIT_CLAIM_LEDGER_20261017.md` · AI coding assistance used for parts of the code and text · authorship pending collaborator review.